

AVINASH SINGH

Software Development Engineer

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SUMMARY

Software Development Engineer specializing in front-end engineering with React and TypeScript, and in building custom Power BI data-visualization components with D3.js and the Power BI Visuals API. Experienced end-to-end — from RESTful API integration and WordPress builds to test-automation platforms and close collaboration with UX/UI teams — with additional hands-on experience shipping real-time, full-stack side projects including multiplayer games and internal tools.

SKILLS

Languages: JavaScript, TypeScript, HTML5, CSS3

Front-End: React.js, Redux, Tailwind CSS, Material UI, Bootstrap, Responsive / Mobile-First Design

Back-End: Node.js, Express.js, RESTful APIs, Socket.IO

Data Visualization & BI: Power BI Custom Visuals, Power BI Visuals API, D3.js, Tableau-to-Power BI Conversion

CMS: WordPress

Database: MongoDB, PostgreSQL, SQLite

Tools & Testing: Git/GitHub, Postman, Docker, JMeter, k6, Locust

EXPERIENCE

BI Hub Solution

Sept 2024 – Sept 2025

Software Development Engineer — Remote

- Developed custom Power BI visuals using TypeScript and D3.js, integrated with the Power BI Visuals API to enhance interactive data visualization.
- Implemented interactive Power BI visuals with dynamic, real-time data updates.
- Built and maintained company websites and landing pages using WordPress.
- Designed and implemented RESTful APIs to strengthen front-end/back-end integration, improving system performance and reliability.
- Partnered closely with the UX/UI team to deliver responsive designs and a consistent user experience.
- Conducted unit testing and debugging to ensure the stability and reliability of shipped features.

Projects Worked On: Load Testing, Tableau-to-Power BI Conversion, Decision Pulse AI, Pulse QA (Front End), Cohort Analysis, Sankey Chart, Supply Chain, Promo Projection Chart, Price Elasticity Heatmap.

Pulse QA

- Built the Pulse QA test-automation platform: a React (Material UI, dnd-kit, CodeMirror) frontend for drag-and-drop test-case building and script editing.
- Cross-Platform Execution: Paired the frontend with dedicated Java-based backend, web, and mobile execution engines so tests run consistently across platforms and devices.
- AI Test Assistant: Built a chat-based AI assistant (FastAPI + MongoDB) that helps analyze test runs and surface corrective actions alongside execution.
- Enabled parallel test execution and CI/CD pipeline integration to reduce total test-cycle time and automate testing during development and deployment.

Load Testing

- Built a full-stack Load & UI Testing platform (FastAPI + MongoDB backend, React/Vite + ApexCharts dashboard) to configure, run, and monitor performance tests from the browser.
- Auto-generated and ran Locust load-test scripts per project/script configuration asynchronously in the background, with retry and stop controls.
- Parsed and stored per-endpoint metrics (request count, failure count, min/median/average/max response time, requests/sec) in MongoDB and visualized them on a reporting dashboard.
- Extended the platform with Selenium- and Katalon-driven UI regression testing, plus JWT-based user, project, and invite management.

Decision Pulse AI

- Natural Language Querying: Implemented a chat/prompt-based interface to generate charts and insights from data using NLP.
- Self-Service Dashboarding & Auto Insights: Enabled drag-and-drop dashboard creation and integrated ML models to auto-detect anomalies, trends, and outliers in datasets.
- Custom KPI Recommendations: Suggested relevant KPIs and visualizations based on dataset schema and business domain.
- Embedded Analytics: Provided an iFrame/SDK for embedding dashboards into web apps and portals.

Cohort Analysis (Custom Power BI Visual)

- Built a custom Power BI visual (TypeScript, D3.js, Power BI Visuals API) to track user retention and engagement trends by cohort, with configurable groupings and color scales in the formatting pane.

Sankey Chart (Custom Power BI Visual)

- Developed a custom Sankey diagram visual for Power BI, using D3.js for flow rendering and the Power BI Visuals API for data binding, to visualize flow and distribution across categories and process stages.

Supply Chain (Custom Power BI Visual)

- Built a custom Power BI visual to track and visualize supply chain metrics for embedded operational dashboards, with interactive filtering and tooltips for drilling into specific stages.

Promo Projection Chart (Custom Power BI Visual)

- Created a custom Power BI visual (TypeScript, Power BI Visuals API) to project outcomes of promotional and marketing activities — sales, engagement, and brand awareness — for business planning.

Price Elasticity Heatmap (Custom Power BI Visual)

- Developed a heatmap visual (TypeScript, Power BI Visuals API) illustrating price elasticity of demand, helping business teams see how price changes affect demand at a glance.

Grapple Tech

Oct 2024 – Sep 2024

Software Engineer — Remote

- Built a personal portfolio website using React.js with a responsive, mobile-first design.
- Showcased detailed project descriptions with live demo and GitHub links.
- Integrated a contact form using EmailJS for direct communication with employers.

Ramy Infotech

April 2021 – Nov 2021

Software Engineer — Noida, Uttar Pradesh

- Implemented responsive web design so applications worked well across devices and screen sizes.
- Optimized front-end performance by improving load times and reducing resource consumption.
- Built responsive, interactive user interfaces using HTML, CSS, and JavaScript.
- Designed APIs for communication between front-end and back-end systems.

PERSONAL PROJECTS

Question Game (Fake Answer Party) — Full-Stack Multiplayer Game

- Built a full-stack, real-time multiplayer bluffing party game with React (Vite) on the front end and Node.js/Express + Socket.IO on the back end.
- Designed multiple game modes (Weird Facts, Fake News, Logo Challenge, World Trivia, and more) with a server-authoritative flow: create room → join → submit → vote → results.
- Used PostgreSQL/Supabase for the question bank and player profiles, and built an AI-assisted question generator to help scale it; deployed front end on Vercel and back end on Railway/Render.

ChorKaun — Real-Time Multiplayer Card Game

- Built a real-time, 4-player web version of the classic card game Raja-Mantri-Chor-Sipahi using React (Vite) on the front end and Express + Socket.IO + SQLite on the back end.
- Implemented room-based multiplayer with shareable 6-digit room codes, rejoin-after-disconnect support, and a persistent SQLite-backed leaderboard.
- Added an in-game push-to-talk voice feature and public matchmaking; containerized with Docker and deployed live on Render.

Weather App

- Developed a responsive weather application using React.js for real-time weather updates, with error handling for invalid searches, API failures, and missing data.
- Designed a mobile-first, responsive UI with search suggestions, location-based weather, and unit toggling.

Currency Converter

- Built an interactive multi-currency converter using React.js with a real-time exchange-rate API and input validation for currency selections.
- Designed a fully responsive UI using CSS and Flexbox for cross-device compatibility.

Doctor Management System

- Built a centralized backend using Express (Node.js) with ES6, and a React.js front end, documented and tested via Postman.
- Integrated Twilio and Microsoft Teams for appointment communication and notifications.

EDUCATION

Lovely Professional University — Phagwara, Punjab

2017 – 2021

Bachelor of Technology in Computer Science and Engineering